Arch Linux + Niri + Zsh: The Ultimate Dotfiles Blueprint

Comprehensive System Installation & Configuration Documentation

This document serves as the absolute, 100% accurate record of everything the `install.sh` script does, every package it installs, and every configuration it applies to transform a blank Arch Linux machine into your bespoke Niri Wayland setup.

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1. The Core Philosophy

We successfully stripped away the bloated, pre-configured setup from the "Maciejonos" dotfiles (which relied on Hyprland, X11 legacy tools, and hardcoded user paths). We retained the beautiful visual aspects (Matugen dynamic theming, Elephant lua menus, Waybar, Walker) but rebuilt the foundation entirely around **Niri** (the scrollable tiling window manager), **Zsh**, and **Nvidia Wayland native compatibility**.

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2. The Exact Package Manifest (Pacman & AUR)

The script installs `yay` from source, then batch-installs the following meticulously vetted packages:

Hardware & Core System Utilities: `zram-generator`, `smartmontools`, `hdparm`, `ufw`, `nss-mdns`, `xpadneo-dkms`, `wacomtablet`, `sbctl`

Base & Development Toolchain: `base-devel`, `git`, `docker`, `docker-compose`, `npm`, `python-pip`, `python-setuptools`, `python312`, `python-pynput`, `pyenv`, `jdk17-openjdk`, `jre8-openjdk`, `clang`, `cmake`, `ninja`, `android-platform`, `android-sdk`, `android-sdk-build-tools`, `android-sdk-cmdline-tools-latest`, `android-sdk-platform-tools`

Shell & CLI Power Tools: `zsh`, `starship`, `zoxide`, `fzf`, `eza`, `bat`, `fd`, `ripgrep`, `jq`, `pv`, `yazi`, `trash-cli`, `unzip`, `wget`, `ncdu`, `tree`, `btop`, `htop`, `fastfetch`, `gum`, `mediainfo`

Terminal Emulators & Web Browsers: `wezterm`, `chromium`, `firefox`

File Managers: `nautilus`, `yazi`, `nautilus-open-any-terminal`

Editors & IDEs: `neovim`, `lazygit`, `visual-studio-code-insiders-bin`, `clion`, `datagrip`, `clion-jre`, `datagrip-jre`

Window Manager, Wayland Base & Theming: `niri`, `fuzzel`, `waybar`, `mako`, `dunst`, `swayosd`, `polkit-gnome`, `gnome-keyring`, `xdg-desktop-portal-gtk`, `xdg-utils`, `satty`, `sunsetr`, `tinte`, `walker`, `xar`, `qt5ct`, `qt6ct`, `qt6-wayland`, `matugen`, `swww`

Media, Audio & Screen (Screenshot, Images, Video): `grim`, `slurp`, `wl-clipboard`, `wl-paste`, `imv`, `mpv`, `vlc`, `yt-dlp`, `ffmpeg`, `imagemagick`, `loupe`, `gpu-screen-recorder`, `handbrake-cli`, `tesseract`, `tesseract-data-eng`, `pipewire`, `wireplumber`, `pipewire-pulse`, `pipewire-alsa`, `pipewire-jack`, `gst-plugin-pipewire`, `pavucontrol`, `pamixer`, `wiremix`, `playerctl`, `libpulse`

Network & Bluetooth: `network-manager-applet`, `networkmanager`, `bluez`, `bluez-utils`, `blueman`, `bluetui`, `iwd`, `sshfs`

System & File Utilities: `gvfs`, `gvfs-dnssd`, `udiskie`, `pacman-contrib`, `brightnessctl`, `power-profiles-daemon`, `plocate`, `vmware-workstation`, `vmware-keymaps`, `cups`, `hplip`, `system-config-printer`

Fonts & Themes: `noto-fonts`, `noto-fonts-emoji`, `ttf-cascadia-mono-nerd`, `ttf-jetbrains-mono-nerd`, `bibata-cursor-theme`, `gnome-themes-extra`, `otf-font-awesome`, `yaru-icon-theme`

Gaming, Launchers & Flatpaks: `steam`, `steam-devices`, `heroic-games-launcher-bin`, `modrinth-app-bin`, `pandora-launcher-bin`, `r2modman-bin`, `balatro-mod-manager-bin` *(Flatpaks)*: `flathub`, `com.usebottles.bottles`, `org.vinegarhq.Sober`, `org.kde.kfind`

Elephant Suite (Custom Menus): `elephant`, `elephant-calc`, `elephant-clipboard`, `elephant-desktopapplications`, `elephant-files`, `elephant-menus`, `elephant-providerlist`, `elephant-symbols`, `elephant-todo`

Other Applications: `discord`, `spotify`, `libreoffice-fresh`, `onlyoffice-bin`, `keepassxc`, `ollama`, `localsend-bin`, `kdeconnect`, `kicad`, `wireshark-qt`, `krita`, `aniworld-cli`, `claude-code`, `scilab-bin`, `teams-for-linux-bin`, `bambustudio-nvidia-bin`, `github-desktop-bin`, `beets`, `picard`, `strawberry`, `high-tide`, `impala`, `input-leap-bin`, `winboat-bin`

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3. Deep Hardware Configuration: Nvidia & Wayland

If an Nvidia GPU is detected (`lspci`), the script fundamentally alters the boot and Wayland environment to prevent tearing, black screens, and cursor glitches:

1. **Environment Variables:** Creates `~/.config/environment.d/10-nvidia.conf` injecting `LIBVA_DRIVER_NAME=nvidia`, `GBM_BACKEND=nvidia-drm`, `__GLX_VENDOR_LIBRARY_NAME=nvidia`, and `WLR_NO_HARDWARE_CURSORS=1`.
2. **Early KMS (Kernel Mode Setting):** Writes `options nvidia_drm modeset=1 fbdev=1` to `/etc/modprobe.d/nvidia.conf` (fixing TTY freezing on Wayland).
3. **Initramfs Rebuild:** Injects `nvidia`, `nvidia_modeset`, `nvidia_uvm`, and `nvidia_drm` directly into `/etc/mkinitcpio.conf` and runs `mkinitcpio -P`. This makes the fix bootloader-agnostic (works on GRUB and systemd-boot).

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4. Dynamic Dotfiles Symlinking

The script safely loops through the repository and symlinks configurations:

- Everything in `config/` (e.g., `niri`, `waybar`, `fastfetch`, `nvim`, `matugen`, `elephant`) is symlinked to `~/.config/`.
- Everything in `home/` (e.g., `.zshrc`, `.profile`, `.gtkrc-2.0`, `.gitconfig`) is symlinked to `~/`.
- Existing files are safely backed up with a `.bak` extension before linking.

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5. Shell Re-Architecture: Zsh + Oh-My-Zsh

We migrated your messy "MyBash" setup into a lightning-fast `~/.zshrc`:

- Sets Zsh as default via `chsh`.
- Sources the **Oh-My-Zsh** framework for Git, Zoxide, and FZF plugins.
- Installs native Arch Linux packages for `zsh-autosuggestions` and `fast-syntax-highlighting` to provide grey autocomplete hints and colored commands.

- **Custom Aliases Kept:** `yayf` (Fuzzy-finding AUR packages with preview), `whatsmyip`, `hb` (Hastebin upload), `rmd` (safe recursive remove).
- **Fastfetch Hook:** Added a custom `add-zsh-hook preexec` that displays your custom boxed Fastfetch logo on terminal startup, but automatically `clear`s the screen the moment you type your first command.

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6. Niri Window Manager & Dynamic Wallpapers

We fully ported your Hyprland keybindings into Niri (`config.kdl`):

- Added `swww-daemon` to Niri's autostart for animated wallpaper transitions.
- Created `~/.config/niri/set-wallpaper.sh`, which receives an image path, changes the wallpaper via `swww`, runs `matugen` to extract the image's dominant colors, and automatically reloads `waybar` and `mako` to instantly apply the new color scheme.
- Fixed the `elephant` Lua wallpaper menu (`config/elephant/menus/dynamic/wallpapers.lua`) to call this new script instead of the hardcoded Hyprland tools.

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7. System & Pacman Tweaks

- **PlatformIO UDEV Rules:** Automatically fetches the latest `99-platformio-udev.rules` from GitHub to `/etc/udev/rules.d/` and reloads `udevadm`, ensuring microcontrollers (Arduino, ESP32) can be flashed immediately.
- **Git Configuration:** If no Git identity is found, the script interactively asks for your Name and Email and configures `.gitconfig` securely without pushing private data to GitHub.
- **Pacman Tweaks:** Edits `/etc/pacman.conf` via `sed` to enable `Color`, inject the `ILoveCandy` Easter egg (Pac-Man eating pills progress bar), and enables `ParallelDownloads = 5` for lightning-fast installs.
- **Multilib Auto-Enable:** Automatically uncomments the `[multilib]` repository in `pacman.conf` (required for installing Steam and 32-bit gaming layers) and runs a database sync.

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8. Systemd Services

Enables essential user and system services for immediate desktop functionality:

- `systemctl --user enable --now walker.service` (App Launcher)
- `systemctl --user enable --now swayosd.service` (Volume/Brightness pop-ups)
- `systemctl --user enable --now elephant.service` (Lua Macro Menus)
- `sudo systemctl enable --now cups.service` (Apple CUPS Printing Daemon)

This blueprint guarantees a 1:1, perfectly customized, Wayland-native Arch Linux environment.